

Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Full Chapter Revision Worksheet

I. Multiple Choice Questions

- EFB stands for:
(A) European Federation of Biotechnology
(B) European Fund for Biology
(C) Environmental Federation of Biology
(D) European Foundation for Bioremediation
- How many essential tools of Recombinant DNA Technology (RDT) are listed in the chapter?
(A) 3
(B) 4
(C) 5
(D) 6
- Plasmids are described as:
(A) Linear, non-replicating DNA
(B) Autonomously replicating, circular, extra-chromosomal DNA
(C) Single-stranded RNA molecules
(D) Protein-based vectors
- Genes that help distinguish recombinants from non-recombinants are called:
(A) Structural genes
(B) Selectable markers
(C) Housekeeping genes
(D) Promoter genes
- The Ti plasmid is associated with which organism?
(A) *Agrobacterium tumefaciens*
(B) *Escherichia coli*
(C) *Bacillus subtilis*
(D) *Thermus aquaticus*
- Which enzyme is used to break open a bacterial cell wall during DNA isolation?
(A) Cellulase
(B) Chitinase
(C) Lysozyme
(D) Protease
- PCR stands for:
(A) Polymerase Chain Reaction
(B) Protein Coding Region
(C) Plasmid Cloning Reaction
(D) Primer Chain Replication
- Introduction of recombinant DNA into bacterial cells is typically achieved by:
(A) Transformation
(B) Photosynthesis induction
(C) Meiosis
(D) Conjugation only

II. Fill in the Blanks

- Genetic engineering includes creation of recombinant DNA, gene cloning, and _____.
- Restriction enzyme recognition sequences are _____ in nature.
- In pBR322, _____ and tetR serve as selectable marker genes.
- Insertion of foreign DNA that disrupts a gene's function is called _____.
- The Ti plasmid used as a plant vector comes from _____.
- Chilled ethanol is added at the final step of DNA isolation to _____ purified DNA.
- Taq polymerase, used in PCR, is isolated from _____.
- Downstream processing includes separation, purification, and _____ of a crude product.

III. Match the Following

Column A	Column B
1. EcoRI	a. Alternate selectable marker
2. ori	b. High-velocity particle bombardment
3. β -galactosidase marker	c. Derived from E. coli, strain R
4. Biolistics	d. Keeps cells in log/exponential phase
5. Taq polymerase	e. Controls vector copy number
6. Continuous culture	f. Thermostable enzyme from <i>Thermus aquaticus</i>

IV. True / False

1. Hybridisation transfers only the desired gene without any unwanted genes. [True / False]
2. Sticky ends allow DNA fragments cut with the same restriction enzyme to be joined together. [True / False]
3. Blue colonies in blue-white screening represent successful recombinants. [True / False]
4. A competent host is a cell capable of taking up vector DNA. [True / False]
5. PCR requires only a single primer complementary to the target DNA. [True / False]
6. Continuous culture keeps cells in their log/exponential phase for higher yield. [True / False]

V. 2 Marks Questions

1. Explain the three basic steps involved in genetically modifying an organism.

2. Describe how sticky ends are formed and why they are essential for constructing recombinant DNA.

VI. 3 Marks Questions

1. Explain how antibiotic resistance genes are used to distinguish recombinants from non-recombinants.

2. Describe the three steps of a PCR cycle and the role of Taq polymerase.

VII. 5 Marks Questions

1. Explain the complete process of Recombinant DNA Technology (RDT), from the isolation of genetic material to obtaining the foreign gene product, in the correct sequential order.
